

Course Code	Name of the Course				
216U03C303	Digital Logic Design				
Teaching Scheme	TH	P	TUT	Total	
	03	-	-	03	
Credits Assigned	03	-	-	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	-	20	30	50	100

Course pre-requisites:

- NIL

Course Objectives:

Digital systems have a prominent role in everyday life. To understand the operation of digital systems, it is necessary to have a basic knowledge of digital circuits and their logical functions. The objective of this course is to familiarize the student with fundamental principles of digital design. It provides coverage of classical hardware design for both combinational and sequential logic circuits.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand fundamentals of number systems, codes and arithmetic operations.
- CO2.** Minimize logic functions using different techniques and represent using logic gates.
- CO3.** Implement combinational logic circuits and their applications using various devices.
- CO4.** Design and implement sequential logic circuits and their applications.
- CO5.** Understand principle of data converters.

Detailed Syllabus

Module No.	Unit No.	Contents	Hrs.	CO
1	Fundamentals of digital design			
	1.1	Number systems and codes: Introduction to digital systems, binary, octal, hexadecimal number systems, signed and unsigned numbers BCD code, Gray code, weighted and non-weighted codes, self-complementary codes and Error detecting code	05	CO1
	1.2	Binary arithmetic- Addition, subtraction using 1's and 2's complement, multiplication, division and BCD Arithmetic		
2	Combinational Logic Circuits			
	2.1	Logic Gates: Review of Basic logic Gates, universal gates, Minimization of logical expression using Boolean Functions. Combinational logic representation using truth table, and standard SOP, POS form	18	CO2, CO3
	2.2	Minimization techniques: Use of Boolean theorem, K- map up to five variables, Quine-McClusky method, realization using standard and universal gates		
	2.3	Arithmetic circuits: Adder, subtractor, controlled adder/ subtractor, BCD adder		
	2.4	Use Of MSI Devices: Multiplexer, de-multiplexer, decoder, encoder, comparator, multiplexer tree and decoder tree. Boolean functions implementation using above devices.		
	2.5	Logic Families: Introduction to logic families, characteristics of digital ICs, transfer characteristics and comparison of TTL and CMOS families.		
	2.6	Programmable logic devices: Introduction, realization of logic expressions with PROM, PLA and PAL # Self-learning: priority encoder, 7 segment decoders		
3	Sequential Logic Circuits			
	3.1	Flip flops (FF): SR, JK, T, D and master slave flip flops, Truth table and excitation table characteristics equation, conversion of flip flops	10	CO4
	3.2	Counter: Asynchronous and synchronous counter, Design of mod counters, Timing diagram		
	3.3	Shift registers: shift left and shift right Registers, bi-directional shift register, applications, ring counter, Johnson counter		
		# Self-learning: Designing with MSI devices (ICs 7490, 7492, 74163, 74194 etc.)		

4	Finite state Machines		08	CO4
	4.1	Finite state machines: Moore and Mealy machine, block diagram, state diagram, analysis of clocked synchronous state machine		
	4.2	Synchronous state machine design: P.S –N.S. table, reduction using implication chart method, state assignments, design using flip flops, up/down counter, skipping state counter, sequence detector and simple FSM design problems		
5	Data Convertors		04	CO5
	5.1	Data converters: A to D Conversion techniques, Dual slope ADC, Ramp ADC, Successive approximation ADC D to A Conversion techniques- R - 2R ladder network		
Total			45	

Self Learning: Students should prepare all self-learning topics on their own. Self learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA and or laboratory experiments.

Reference Books*

Sr No	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1	Morris Mano and Michael D. Ciletti	<i>Digital Design: With an Introduction to Verilog HDL</i>	Pearson Education, India	5 th Edition 2013
2	A.P. Malvino and D.P. Leach	<i>Digital Principles and Applications</i>	McGraw-Hill Education, India	10 th Edition 2015
3	R.P. Jain	<i>Modern Digital Electronics</i>	Tata McGraw Hill Education, India	4 th Edition 2015
4	John M Yarbrough	<i>Digital logic: Applications and design</i>	Thomson Brooks/ Cole, India	India Edition 2006